



DESIGN AND IMPLEMENTATION OF AI DISEASE PREDICTION AND MEDICINE RECOMMENDATION SYSTEM USING MACHINE LEARNING

A MINI PROJECT REPORT

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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

**P.S.V. COLLEGE OF ENGINEERING AND TECHNOLOGY
KRISHNAGIRI**

ABSTRACT

The application of Artificial Intelligence (AI) and Machine Learning (ML) in healthcare has transformed the way diseases are detected and diagnosed. This mini project focuses on the design and implementation of an **AI-based Disease Prediction and Medicine Recommendation System** that predicts potential diseases based on user-input symptoms and suggests corresponding medicines along with dosage information. The system is built using Python, with a **Random Forest Classifier** trained on a healthcare dataset containing symptoms, diseases, and treatments. Symptom data is preprocessed and encoded using the **MultiLabelBinarizer** technique, while diseases are label-encoded to improve model accuracy. The web interface is developed using **Streamlit**, styled with custom CSS to ensure a clean, intuitive, and user-friendly experience. Users can input details such as age, weight, and up to four symptoms, after which the system processes the data and predicts the most probable disease. It then provides medicine recommendations along with proper dosage guidelines. This system is particularly beneficial in rural or remote areas where immediate access to healthcare professionals is limited, offering preliminary guidance before professional consultation. While it does not replace certified medical advice, the system provides fast and accurate predictions, making it a valuable supplementary tool in healthcare. Potential future enhancements include adding a voice-based symptom input feature, storing patient history in a database, creating user authentication for doctors and patients, and deploying the application on a cloud platform for global access.

Hardware and Software Requirements

1. Hardware Requirements

- **Processor:** Intel Core i3 or equivalent (Minimum), Intel Core i5/i7 or equivalent (Recommended)
- **RAM:** 4 GB (Minimum), 8 GB or higher (Recommended)
- **Storage:** 500 MB free space (Minimum), 1 GB or more (Recommended)
- **Display:** 1024 × 768 resolution (Minimum), 1920 × 1080 resolution (Recommended)
- **Input Devices:** Keyboard and Mouse
- **Internet Connection:** Basic internet (for installing packages and APIs), Stable broadband recommended

2. Software Requirements

- **Operating System:** Windows 10/11, Linux, or macOS
- **Programming Language:** Python 3.8 or higher
- **Framework:** Streamlit for frontend user interface
- **Libraries:** pandas, scikit-learn, numpy, MultiLabelBinarizer, LabelEncoder, joblib
- **Code Editor:** Visual Studio Code or PyCharm
- **Dataset Format:** CSV (updated_disease_dataset.csv)
- **Version Control (optional):** Git and GitHub
- **Browser:** Google Chrome, Microsoft Edge, or Firefox
- **Deployment (optional):** Streamlit Cloud, Heroku, or AWS